Project Title:

FarmChainX – AI-Powered Farm Product Traceability System

Project Statement:

FarmChainX is a blockchain-inspired, AI-enabled platform that ensures transparent traceability of agricultural products from farm to fork. The system uses AI for anomaly detection, predictive quality scoring, and integrates QR-based tracking for stakeholders such as farmers, distributors, and consumers. Built using Spring Boot, Angular 19, and MySQL, it offers role-based dashboards for real-time product visibility, ensuring food safety, authenticity, and quality assurance in the agriculture supply chain.

Module-Wise Implementation with Features & Expected Output

1. User & Role Management

Roles: Farmer, Distributor, Retailer, Consumer, Admin

Services:

- Register/Login
- Role-based navigation and permissions

Expected Output:

- Login/Register page with dropdown for role
- Separate dashboards (e.g., Farmer sees upload form; Consumer sees product info)

2. Farm Product Upload & Metadata Collection (Farmer)

Services:

- Add new product (crop, fruit, etc.)
- Capture metadata (crop type, soil type, pesticides used, harvest date)
- Upload image & GPS location

Expected Output:

- Form with fields and file uploader
- Location capture using Google Maps API
- Confirmation message upon submission

3. QR Code Generation & Assignment

Services:

- Generate unique QR for each product batch
- Encode batch metadata in QR
- Assign QR to product ID

Expected Output:

- Downloadable QR image with tracking link
- QR preview page with share option

4. AI Quality Scoring Module

Services:

- Upload product image for AI-based quality prediction
- Use ML model to classify grades (A, B, C)

- Show confidence % of prediction

Expected Output:

- Image preview + Predicted quality
- Progress bar showing prediction accuracy

5. Supply Chain Tracking & Handover (Distributors & Retailers)

Services:

- Accept product batch via QR scan
- Update transport, storage, and handover logs
- View product history timeline

Expected Output:

- Timeline UI showing chain of custody
- Log update page for tracking events

6. Consumer Product Verification Portal

Services:

- Scan QR to view full product journey
- View quality rating, location, and farmer info
- Submit feedback/rating

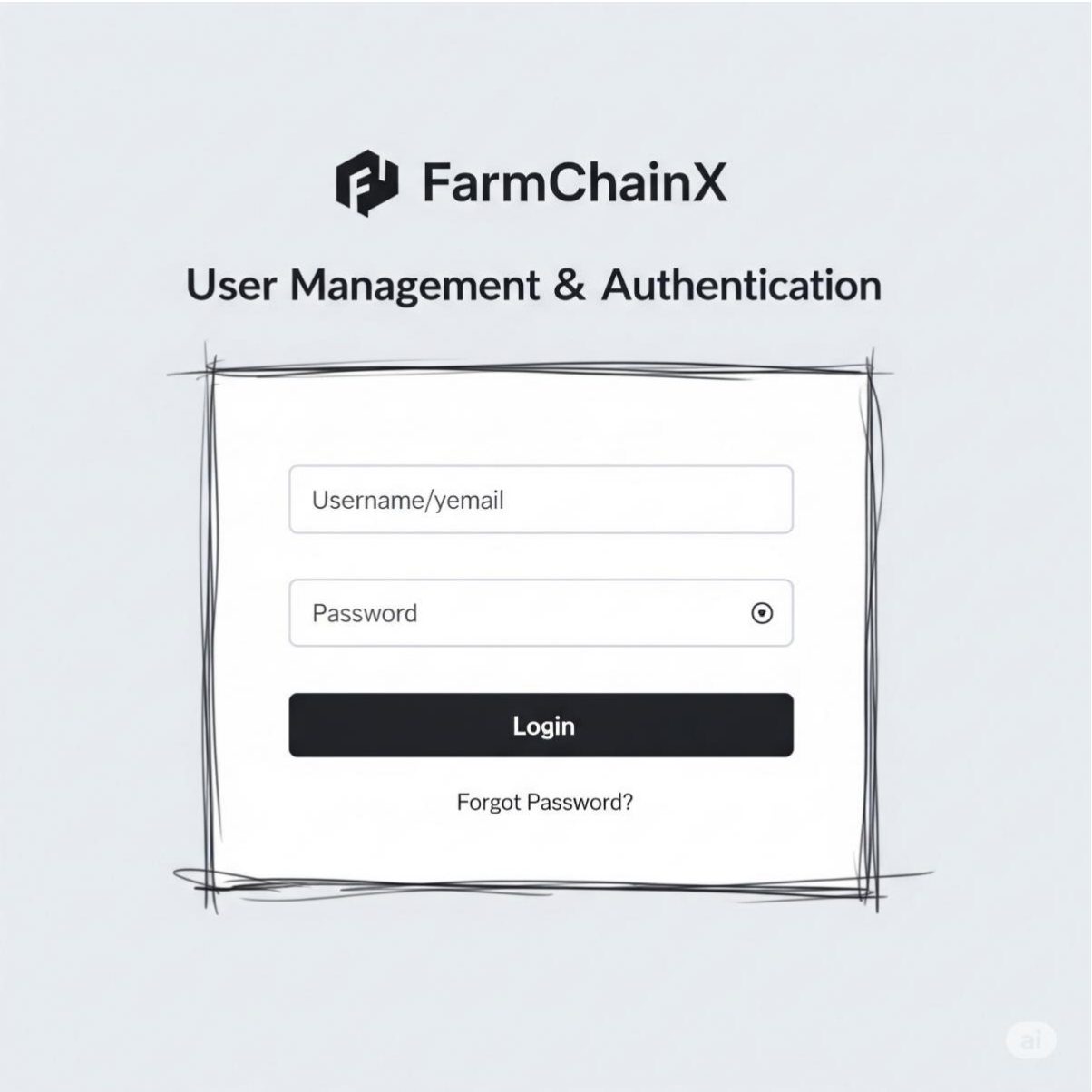
Expected Output:

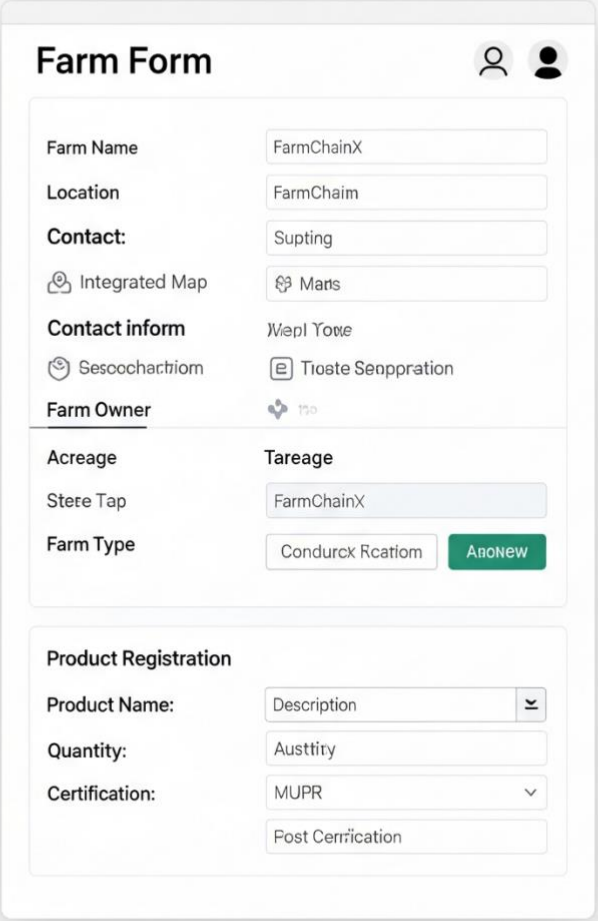
- Journey visual with maps and timeline
- Product details with rating & comment box

Layer	Technology
Frontend	Angular 19, QR Scanner
Backend	Spring Boot, REST APIs, Spring Security
AI Module	Python Flask API / TensorFlow model
Database	MySQL with Hibernate JPA
Extras	QR Code Generator, Google Maps API
DevOps	Postman, GitHub
Charts	Chart.js / ApexCharts / Recharts

Other APIs	Email (SMTP), WebSocket for live updates

Expected outputs:

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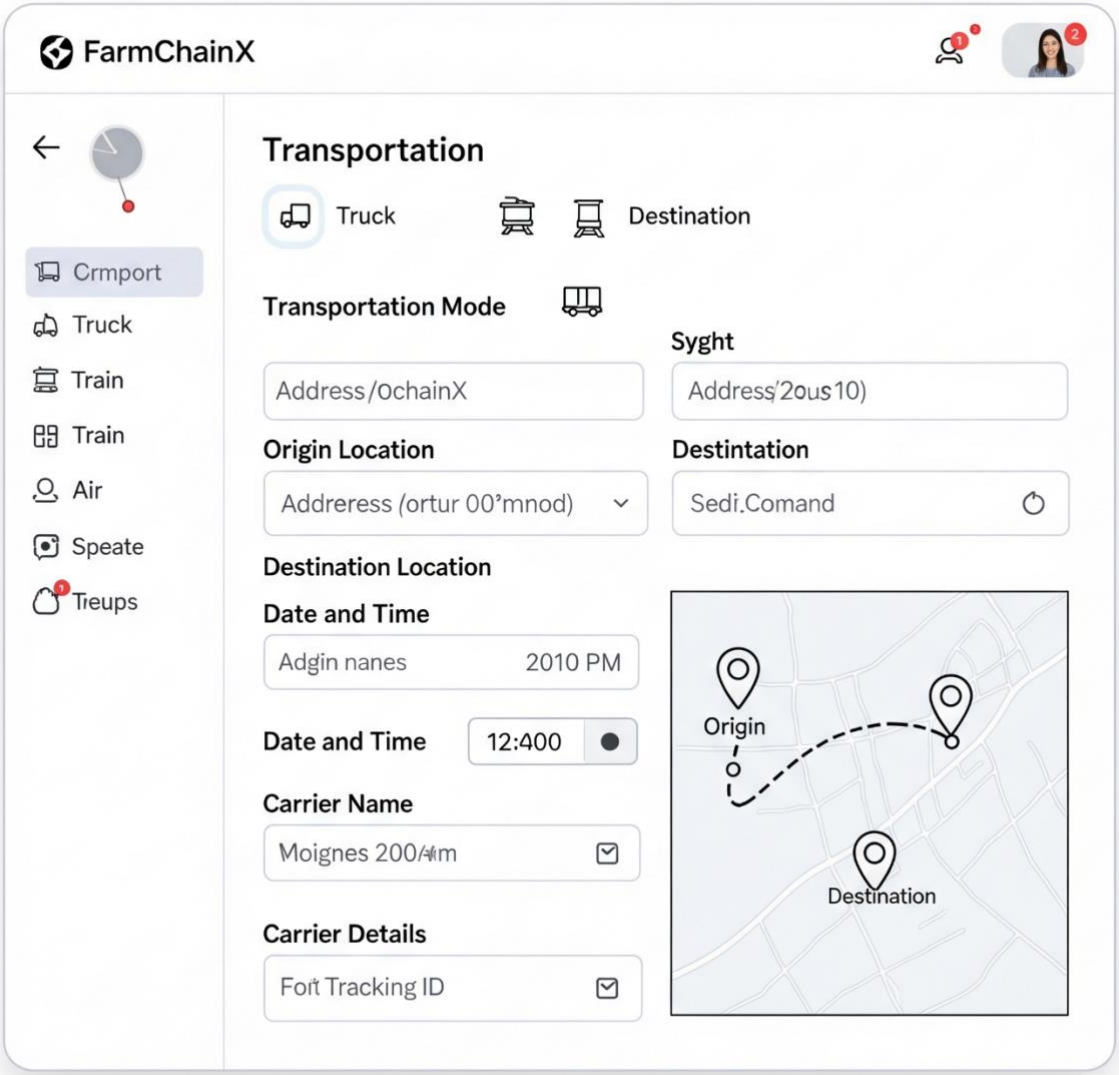
FarmChainX
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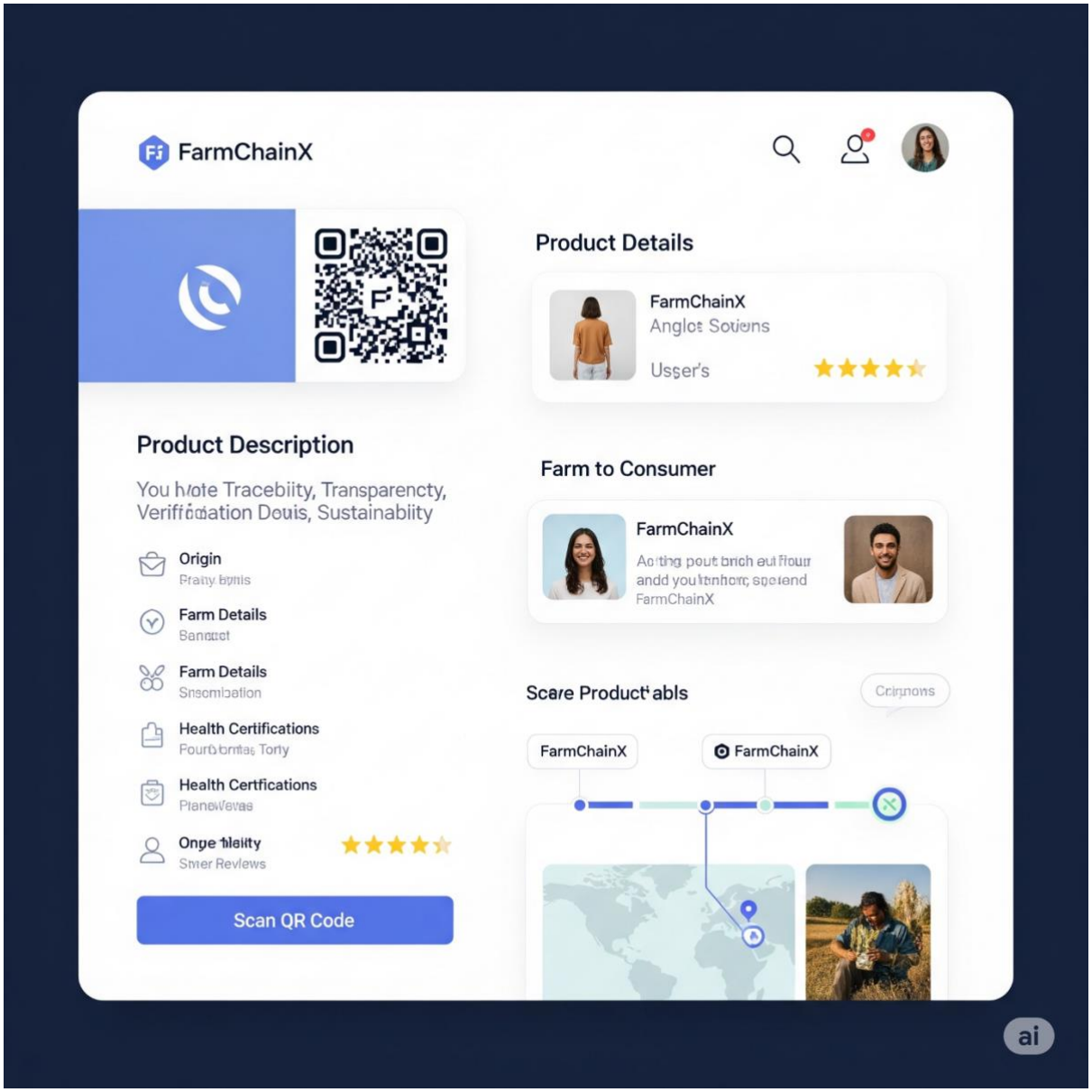
FarmChainX
Product Details
FarmChainX
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Product Description
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Farm Details
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AI-Powered Analytice and Reporting
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